# Computer Networks Project Part 1

## Spring Semester 25-26


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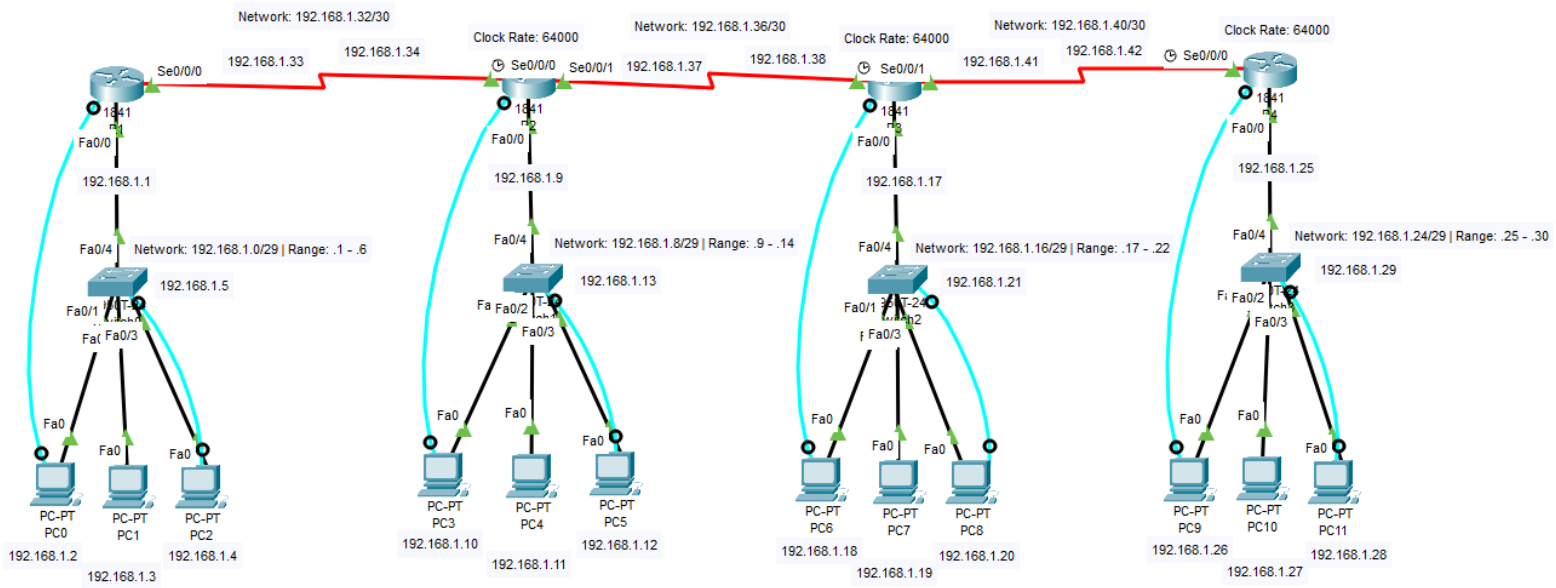
## 1) Network Topology Diagram:



## 2) Addressing Table:

| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| **R1** | Fa0/0 | 192.168.1.1 | 255.255.255.248 | N/A |
| | Se0/0/0 (DTE) | 192.168.1.33 | 255.255.255.252 | N/A |
| **R2** | Fa0/0 | 192.168.1.9 | 255.255.255.248 | N/A |
| | Se0/0/0 (DCE) | 192.168.1.34 | 255.255.255.252 | N/A |
| | Se0/0/1 (DTE) | 192.168.1.37 | 255.255.255.252 | N/A |
| **R3** | Fa0/0 | 192.168.1.17 | 255.255.255.248 | N/A |
| | Se0/0/0 (DCE) | 192.168.1.38 | 255.255.255.252 | N/A |
| | Se0/0/1 (DTE) | 192.168.1.41 | 255.255.255.252 | N/A |
| **R4** | Fa0/0 | 192.168.1.25 | 255.255.255.248 | N/A |
| | Se0/0/0 (DCE) | 192.168.1.42 | 255.255.255.252 | N/A |
| **S1** | VLAN 1 | 192.168.1.5 | 255.255.255.248 | 192.168.1.1 |
| **S2** | VLAN 1 | 192.168.1.13 | 255.255.255.248 | 192.168.1.9 |
| **S3** | VLAN 1 | 192.168.1.21 | 255.255.255.248 | 192.168.1.17 |
| **S4** | VLAN 1 | 192.168.1.29 | 255.255.255.248 | 192.168.1.25 |

| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| **PC0** | NIC | 192.168.1.2 | 255.255.255.248 | 192.168.1.1 |
| **PC1** | NIC | 192.168.1.3 | 255.255.255.248 | 192.168.1.1 |
| **PC2** | NIC | 192.168.1.4 | 255.255.255.248 | 192.168.1.1 |
| **PC3** | NIC | 192.168.1.10 | 255.255.255.248 | 192.168.1.9 |
| **PC4** | NIC | 192.168.1.11 | 255.255.255.248 | 192.168.1.9 |
| **PC5** | NIC | 192.168.1.12 | 255.255.255.248 | 192.168.1.9 |
| **PC6** | NIC | 192.168.1.18 | 255.255.255.248 | 192.168.1.17 |
| **PC7** | NIC | 192.168.1.19 | 255.255.255.248 | 192.168.1.17 |
| **PC8** | NIC | 192.168.1.20 | 255.255.255.248 | 192.168.1.17 |
| **PC9** | NIC | 192.168.1.26 | 255.255.255.248 | 192.168.1.25 |
| **PC10** | NIC | 192.168.1.27 | 255.255.255.248 | 192.168.1.25 |
| **PC11** | NIC | 192.168.1.28 | 255.255.255.248 | 192.168.1.25 |

## 3) Verification Screenshots:

### 3.1) Showing Interface Brief on R1 and R4:

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.1.1     YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/0/0            192.168.1.33    YES manual up                    up
Serial0/0/1            unassigned      YES unset  administratively down down
Serial0/1/0            unassigned      YES unset  administratively down down
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down

R4#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.1.25    YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/0/0            192.168.1.42    YES manual up                    up
Serial0/0/1            unassigned      YES unset  administratively down down
Serial0/1/0            unassigned      YES unset  administratively down down
Serial0/1/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

### 3.2) Showing IP Route on R2 and R3:

```
R2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    11.0.0.0/8 is directly connected, Serial0/0/0
     192.168.1.0/24 is variably subnetted, 6 subnets, 3 masks
S       192.168.1.0/24 [1/0] via 11.0.0.2
S       192.168.1.0/29 [1/0] via 192.168.1.33
C       192.168.1.8/29 is directly connected, FastEthernet0/0
S       192.168.1.16/29 [1/0] via 192.168.1.38
S       192.168.1.24/29 [1/0] via 192.168.1.38
C       192.168.1.36/30 is directly connected, Serial0/0/1


R3#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.1.0/24 is variably subnetted, 6 subnets, 2 masks
S       192.168.1.0/29 [1/0] via 192.168.1.37
S       192.168.1.8/29 [1/0] via 192.168.1.37
C       192.168.1.16/29 is directly connected, FastEthernet0/0
S       192.168.1.24/29 [1/0] via 192.168.1.42
C       192.168.1.36/30 is directly connected, Serial0/0/0
C       192.168.1.40/30 is directly connected, Serial0/0/1
```

### 3.3) Ping From PC0 To PC11:

```
C:\>ping 192.168.1.28

Pinging 192.168.1.28 with 32 bytes of data:

Reply from 192.168.1.28: bytes=32 time=13ms TTL=124
Reply from 192.168.1.28: bytes=32 time=39ms TTL=124
Reply from 192.168.1.28: bytes=32 time=16ms TTL=124
Reply from 192.168.1.28: bytes=32 time=25ms TTL=124

Ping statistics for 192.168.1.28:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 13ms, Maximum = 39ms, Average = 23ms
```

### 3.4) Ping From PC5 To PC6:

```
C:\>ping 192.168.1.18

Pinging 192.168.1.18 with 32 bytes of data:

Reply from 192.168.1.18: bytes=32 time=2ms TTL=126
Reply from 192.168.1.18: bytes=32 time=1ms TTL=126
Reply from 192.168.1.18: bytes=32 time=1ms TTL=126
Reply from 192.168.1.18: bytes=32 time=16ms TTL=126

Ping statistics for 192.168.1.18:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 16ms, Average = 5ms
```

**3.5.1) Configuration Files (R1):**

```
R1#show running-config
Building configuration...

Current configuration : 994 bytes
!
version 12.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
!
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface FastEthernet0/0
ip address 192.168.1.1 255.255.255.248
duplex auto
speed auto
!
interface FastEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
```

```
interface Serial0/0/0
ip address 192.168.1.33 255.255.255.252
!
interface Serial0/0/1
no ip address
clock rate 2000000
shutdown
!
interface Serial0/1/0
no ip address
clock rate 2000000
shutdown
!
interface Serial0/1/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
ip classless
ip route 192.168.1.8 255.255.255.248 192.168.1.34
ip route 192.168.1.16 255.255.255.248 192.168.1.34
ip route 192.168.1.24 255.255.255.248 192.168.1.34
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
End
```

**3.5.2) Configuration Files (R2):**

```
R2#show running-config
Building configuration...

Current configuration : 1008 bytes
!
version 12.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R2
!
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface FastEthernet0/0
ip address 192.168.1.9 255.255.255.248
duplex auto
speed auto
```

```
!
interface FastEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/0/0
ip address 192.168.1.34 255.255.255.252
clock rate 64000
!
interface Serial0/0/1
ip address 192.168.1.37 255.255.255.252
!
interface Serial0/1/0
no ip address
clock rate 2000000
shutdown
!
interface Serial0/1/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
ip classless
ip route 192.168.1.0 255.255.255.248 192.168.1.33
ip route 192.168.1.16 255.255.255.248 192.168.1.38
ip route 192.168.1.24 255.255.255.248 192.168.1.38
!
ip flow-export version 9
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
end
```

**3.5.3) Configuration Files (R3):**

```
R3#show running-config
Building configuration...

Current configuration : 1008 bytes
!
version 12.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R3
!
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface FastEthernet0/0
ip address 192.168.1.17 255.255.255.248
duplex auto
speed auto
```

```
!
interface FastEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/0/0
ip address 192.168.1.38 255.255.255.252
clock rate 64000
!
interface Serial0/0/1
ip address 192.168.1.41 255.255.255.252
!
interface Serial0/1/0
no ip address
clock rate 2000000
shutdown
!
interface Serial0/1/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
ip classless
ip route 192.168.1.0 255.255.255.248 192.168.1.37
ip route 192.168.1.8 255.255.255.248 192.168.1.37
ip route 192.168.1.24 255.255.255.248 192.168.1.42
!
ip flow-export version 9
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
end
```

**3.5.4) Configuration Files (R4):**

```
R4#show running-config
Building configuration...

Current configuration : 1012 bytes
!
version 12.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R4
!
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface FastEthernet0/0
ip address 192.168.1.25 255.255.255.248
duplex auto
speed auto
```

```
!
interface FastEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/0/0
ip address 192.168.1.42 255.255.255.252
clock rate 64000
!
interface Serial0/0/1
no ip address
clock rate 2000000
shutdown
!
interface Serial0/1/0
no ip address
clock rate 2000000
shutdown
!
interface Serial0/1/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
ip classless
ip route 192.168.1.0 255.255.255.248 192.168.1.41
ip route 192.168.1.8 255.255.255.248 192.168.1.41
ip route 192.168.1.16 255.255.255.248 192.168.1.41
!
ip flow-export version 9
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
end
```

**3.5.5) Configuration Files (S1):**

```
Switch#show running-config
Building configuration...

Current configuration : 1128 bytes
!
version 12.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
```

```
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
ip address 192.168.1.5 255.255.255.248
!
ip default-gateway 192.168.1.1
!
!
!
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
!
!
end
```

**3.5.6) Configuration Files (S2):**

```
Switch#show running-config
Building configuration...

Current configuration : 1129 bytes
!
version 12.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
```

```
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
ip address 192.168.1.13 255.255.255.248
!
ip default-gateway 192.168.1.9
!
!
!
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
!
!
end
```

**3.5.7) Configuration Files (S3):**

```
Switch#show running-config
Building configuration...

Current configuration : 1130 bytes
!
version 12.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
```

```
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
ip address 192.168.1.21 255.255.255.248
!
ip default-gateway 192.168.1.17
!
!
!
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
!
!
end
```

**3.5.8) Configuration Files (S4):**

```
Switch#show running-config
Building configuration...

Current configuration : 1130 bytes
!
version 12.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
```

```
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
!
interface GigabitEthernet0/2
!
interface Vlan1
ip address 192.168.1.29 255.255.255.248
!
ip default-gateway 192.168.1.25
!
!
!
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
!
!
end
```

### 3.6) Interface Brief (S1):

```
Switch#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual up                    up
FastEthernet0/4        unassigned      YES manual up                    up
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
FastEthernet0/22       unassigned      YES manual down                  down
FastEthernet0/23       unassigned      YES manual down                  down
FastEthernet0/24       unassigned      YES manual down                  down
GigabitEthernet0/1     unassigned      YES manual down                  down
GigabitEthernet0/2     unassigned      YES manual down                  down
Vlan1                  192.168.1.5     YES manual up                    up
```